Report 5

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I. Introduction

In this experiment, we want to design a lead and lag controller for an flexible joint robot.
A flexible joint robot is a type of robot designed with joints that offer some degree of flexibility, compliance, or elasticity, distinguishing them from rigid robots. These robots incorporate elements like springs, elastic actuators, or specially designed mechanisms to allow controlled movement with some give or deformation at the joints.

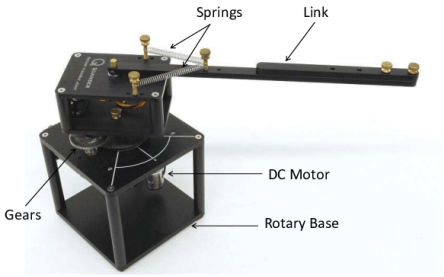


Fig. 1. Flexible Joint Robot

II. Mathematical Modeling

We use the following diagram to derive the mathematical model of the system:

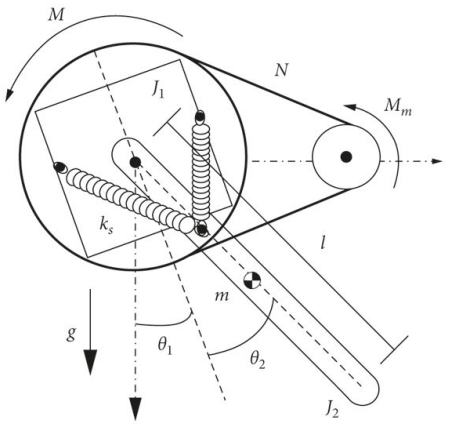


Fig. 2. Flexible Joint Robot Model

Using Lagrange-Hamilton, we arrive at the following equations.

$$J_{eq}\ddot{\theta} + J_{Arm}(\ddot{\theta} + \ddot{\alpha}) = T_{output} - B_{eq}\dot{\theta} \tag{1}$$

$$J_{Arm}(\ddot{\theta} + \ddot{\alpha}) + K_{stiff}\alpha = 0 \tag{2}$$

And we have the torque of a DC motor:

$$T_{output} = \frac{\eta_m \eta_g K_g K_m (V_m - K_g K_m \dot{\theta})}{R_m} \tag{3}$$

The selected state variables are:

$$X = \begin{pmatrix} \theta \\ \dot{\theta} \\ \alpha \\ \dot{\alpha} \end{pmatrix}$$

Then, using the solve command in MATLAB, we solve the above equations and obtain the state-space matrices:

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & \frac{-(K_m K_t n_g n_m K_g^2 + B_{eq} R_m)}{a} & \frac{K_{stiff}}{J_{eq}} & 0 \\ 0 & 0 & 0 & 1 \\ g & \frac{(K_m K_t n_g n_m K_g^2 + B_{eq} R_m)}{a} & \frac{-K_{stiff}(J_{arm} + J_{eq})}{J_{eq} J_{arm}} & 0 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 \\ \frac{K_t K_g \eta_g \eta_m}{J_{eq} R_m} \\ 0 \\ -\frac{K_t K_g \eta_g \eta_m}{J_{eq} R_m} \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix} \qquad D = 0$$

The Lead compensator is used to improve the transient response and has a positive phase, which increases the system's relative stability:

$$G_c(s) = K\frac{s+Z}{s+P}, \quad |P| > |Z|$$

That we use this equation form for design:

$$G_c(s) = K_c \alpha \frac{Ts+1}{\alpha T s + 1} = K_c \frac{s + \frac{1}{T}}{s + \frac{1}{\alpha T}}, \quad 0 < \alpha < 1$$

The pole of the lead controller is always placed to the left of zero, and as the parameter α increases, this pole gets closer to zero. In the practical construction of this controller, there are certain constraints that typically require a minimum value for α, which is generally 0.05. This suggests that the maximum phase that a lead controller can generate is approximately 65 degrees.
$\phi_m$represents the maximum phase that the controller can generate for a given value of α, while ω_m indicates the frequency at which this maximum phase is achieved:

$$\phi_m = PM_{desired} - PM + 5 \tag{4}$$

$$\sin(\phi_m) = \frac{1-\alpha}{1+\alpha} \tag{5}$$

$$\omega_m = \frac{1}{\sqrt{\alpha}T} \tag{6}$$

Thus:

$$\phi_m = 12.5 \qquad \alpha = .6456$$

$$\omega_m = 10.5 \qquad T = 0.1185$$

$$\alpha = \frac{Z}{P} \quad \omega_m = \sqrt{ZP} \tag{7}$$

Thus:

$$Z = 8.367 \quad P = 13.07$$

this is the bode diagram of the system before applying the lead controller:

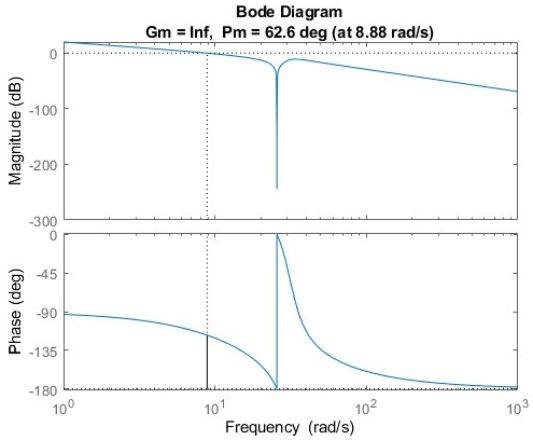


Fig. 3. System Bode Diagram

And this is the bode diagram of the system after applying the lead controller:

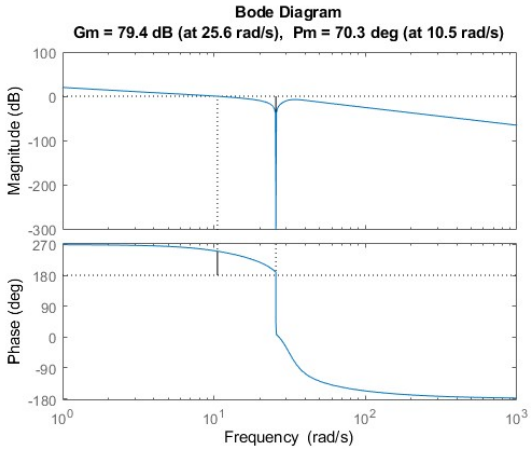


Fig. 4. System Bode Diagram With Lead Controller

III. Simulink

After all we plant the system on simulink. Here is the result:

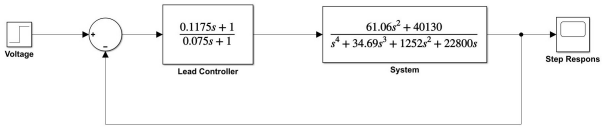


Fig. 5. System With Lead Controller Simulink